

Raspberry Pi 5 Homelab & SSD Protection Cheatsheet

System Administration • Real-Time IO Monitoring • Docker Wear Mitigation • Audio Pipeline Reference

1. Emergency SSD Wear & Write Spike Triage

When your NVMe drive experiences sudden write spikes (e.g., jumping from 0.00 B/s to 12+ MB/s), follow this three-stage triage sequence to identify and neutralize the rogue process.

Stage 1: Catch Real-Time Active Write Loops (iotop)

Use **iotop** to filter out idle processes and display cumulative megabytes-per-second disk write activity.

```
sudo apt update && sudo apt install -y iotop
sudo iotop -o -P
```

Flag Breakdown: **-o** (--only) shows only processes actively reading or writing. **-P** shows process IDs instead of threads, giving clean process names like **python3** (HA) or **postgres** (Immich).

Stage 2: Check Total Block I/O per Docker Container

Use Docker stats to inspect which container has accumulated massive lifetime writes in the **BLOCK I/O** column.

```
docker stats --no-stream --format "table {{.Name}}          {{.CPUPerc}}      {{.MemUsage}}
{{.BlockIO}}"
```

Stage 3: Locate Ballooning Container Log Files

When containers enter rapid crash/restart loops or log flooding (like BlueZ Bluetooth or Tuya cloud rate limits), JSON log files can balloon to gigabytes in minutes.

```
# Show top 5 largest Docker container JSON logs on the NVMe disk
sudo du -ah /var/lib/docker/containers/ | grep -v "/" | sort -rh | head -n 5
```

2. Docker Log Management & Permanent Wear Protection

Never let container logs degrade your NVMe SSD. Use zero-byte truncation for immediate cleanup and apply global daemon log rotation to permanently limit log sizes.

Immediate Emergency Log Truncation (No Container Restart Required)

If a container log hits 500+ MB (e.g., Syncthing or Home Assistant), shrink it back to zero bytes without stopping the container:

```
# Truncate all Docker container JSON logs to 0 bytes safely
sudo truncate -s 0 /var/lib/docker/containers/*/*-json.log

# Or truncate a specific container by ID prefix (e.g., Syncthing 0938100d...)
sudo truncate -s 0 /var/lib/docker/containers/0938100d*/*-json.log
```

Permanent SSD Protection: Global Docker Log Rotation (/etc/docker/daemon.json)

Configure Docker's global daemon settings to cap every container to a maximum of **3 files at 10 MB each** (30 MB total lifetime storage per container).

```
sudo nano /etc/docker/daemon.json
```

Paste the following JSON configuration and restart the Docker daemon:

```
{
  "log-driver": "json-file",
  "log-opts": {
    "max-size": "10m",
    "max-file": "3"
  }
}
```

```
sudo systemctl restart docker
```

IMPORTANT NOTE ON LOG CAPS

Existing containers will continue using old log rules until recreated. To apply rotation rules to currently running containers, recreate them via `docker compose down && docker compose up -d`.

3. Real-Time System Monitoring Cheatsheet (htop & iotop)

Essential key bindings and terminal commands for diagnosing CPU, RAM, Swap, and disk bottlenecks on Debian Trixie / Raspberry Pi OS.

COMMAND / KEYBINDING	ACTION & PRACTICAL USE CASE
htop	Launch interactive CPU/RAM/Process viewer. Displays per-core utilization and memory footprint.
F6 (in htop)	Sort process list. Select PERCENT_CPU or PERCENT_MEM to spot resource hogs.
F9 (in htop)	Kill selected process. Choose signal 15 (SIGTERM) for graceful shutdown or 9 (SIGKILL) to force stop.
F4 (in htop)	Filter processes by string (e.g., type <code>mpd</code> , <code>python3</code> , or <code>node</code>).
sudo iotop -o -P	Show only active IO processes aggregated by main Process ID.
left/right arrows (iotop)	Switch sorting between active DISK READ , DISK WRITE , and IO % .
r (in iotop)	Toggle sorting order between ascending and descending.
free -h	Display available physical RAM, cache, and ZRAM/Swap utilization in readable format.
df -h /var/lib/docker	Check total remaining NVMe disk space available to Docker and system volumes.

4. Home Assistant Audio & Media Pipeline (MPD, BlueZ, PulseAudio)

Commands for troubleshooting the lightweight native audio stack (< 35 MB RAM) routing to your **HAVIT MX701** Bluetooth speaker.

Music Player Daemon (MPD) Command-Line Control (mpc)

```
# Check current playback status, volume level, and active audio error flags
mpc status

# List all local alarm MP3 files inside /var/lib/mpd/music/
mpc ls alarms

# Clear queue, add alarm siren, set volume to 100%, and test playback
mpc clear && mpc add "alarms/nuclear_siren.mp3" && mpc volume 100 && mpc play

# Stop MPD playback immediately
mpc stop
```

Bluetooth Speaker Pairing & Keep-Alive Triage (BlueZ)

```
# Connect directly to HAVIT MX701 Bluetooth speaker MAC address
sudo bluetoothctl connect 17:18:07:A9:99:2D

# Verify connection status and BlueZ audio sink registration
sudo bluetoothctl info 17:18:07:A9:99:2D

# Restart MPD service after Bluetooth pairing or PulseAudio socket changes
sudo systemctl restart mpd
```

Raspotify (Spotify Connect endpoint) Configuration

```
# Edit Raspotify audio backend and speaker name
sudo nano /etc/raspotify/conf

# Ensure LIBRESPOT_NAME="HAVIT MX701 Soundbox" and LIBRESPOT_BACKEND="pulseaudio"

# Restart Raspotify after editing configuration
sudo systemctl restart raspotify
```

5. Service-Specific Crucial Maintenance Stack

Targeted maintenance commands for the core software running on your homelab to prevent database bloat, SSD wear, and connection timeouts.

Home Assistant (HA Core & Logger/Recorder Tuning)

To prevent Home Assistant from writing 40+ GB of database checkpoints and log spam to your SSD, apply these recorder and logger filters in `/home/redwannabil/homeassistant/configuration.yaml`:

```
logger:
  default: warning
  logs:
    # Silence BLE scanner timeout spam and Tuya API rate-limit errors:
    habluetooth.scanner: critical
    tuya_sharing.customerapi: critical

recorder:
  commit_interval: 30
  exclude:
    domains:
      - uptime_kuma
      - systemmonitor
      - ping
```

```
# Test HA YAML configuration before restarting
docker exec -it homeassistant python -m homeassistant --script check_config -c /config
```

Syncthing (Folder Hashing & Database Maintenance)

```
# Check Syncthing container log for permission or sync loop errors
docker logs --tail 50 syncthing

# Truncate Syncthing JSON log if log flooding occurs during initial scans
sudo truncate -s 0 /var/lib/docker/containers/0938100d*/*-json.log
```

Immich Server & PostgreSQL Database Maintenance

```
# Inspect running background jobs and transcode tasks
docker logs --tail 30 immich_server

# Perform PostgreSQL VACUUM ANALYZE to reclaim dead database tuples on SSD
docker exec -it immich_postgres psql -U postgres -d immich -c "VACUUM ANALYZE;"
```

Filebrowser Quantum & Nextcloud

```
# Restart Filebrowser if indexing hangs
docker restart filebrowser_quantum

# Run Nextcloud cron background tasks manually to clear queue
docker exec -u 33 -it nextcloud-app-1 php /var/www/html/cron.php
```

6. Master Quick-Reference Command Table

Complete single-page summary of essential Raspberry Pi 5 homelab commands.

TASK / SCENARIO	TERMINAL COMMAND
Find Top SSD Disk Writers	<code>sudo iotop -o -P</code>
Check Docker Lifetime Block I/O	<code>docker stats --no-stream</code>
Find Top 5 Giant Docker Logs	<code>sudo du -ah /var/lib/docker/containers/ grep -v "/" sort -rh head -n 5</code>
Emergency Log Wipe (All Containers)	<code>sudo truncate -s 0 /var/lib/docker/containers/*/ *-json.log</code>
Restart Global Docker Service	<code>sudo systemctl restart docker</code>
Test Alarm Playback (MPD)	<code>mpc clear && mpc add "alarms/nuclear_siren.mp3" && mpc volume 100 && mpc play</code>
Reconnect HAVIT MX701 Bluetooth	<code>sudo bluetoothctl connect 17:18:07:A9:99:2D</code>
Check RAM & Swap/ZRAM Usage	<code>free -h</code>
Reclaim Immich DB Space	<code>docker exec -it immich_postgres psql -U postgres -d immich -c "VACUUM ANALYZE;"</code>
Edit Home Assistant Config	<code>sudo nano /home/redwannabil/homeassistant/configuration.yaml</code>